

Economics**Higher level****Paper 3****SME Mark Scheme — Paper 3 Advanced**

1. Use the question-specific mark scheme together with the mark bands. Award up to the maximum marks as indicated.
2. Own-figure-rule (OFR) applies. Reward appropriate use of an incorrect figure in subsequent parts.
3. Candidates may be penalised once for rounding errors, missing units or failure to give answers to 2 d.p.

Question 1 — UK grocery oligopoly & behavioural economics**1. (a) (i) Define the term *oligopoly*. [2]**

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>Vague definition.</i> The idea that an oligopoly is a market dominated by a few large firms.	1
2	<i>Accurate definition.</i> An understanding that an oligopoly is a market structure in which a small number of large, interdependent firms dominate the market, typically with significant barriers to entry, product differentiation, and the potential for collusive or strategic behaviour.	2

(ii) Calculate the four-firm concentration ratio (CR4). [2]

CR4 = 27.5 + 15.8 + 13.7 + 8.0 = **65.0%**.

Valid working [1]. Correct answer with % sign [1].

(iii) Explain, using a kinked-demand-curve diagram, why firms in an oligopoly may exhibit price rigidity. [4]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>There is a correct diagram OR an accurate written response.</i> For a diagram showing a demand curve with a kink at the prevailing price with a discontinuity in MR OR an explanation of the asymmetric assumption — rivals match price cuts but not price rises [1], so a firm has no incentive to change price [1].	1-2

2	<i>There is a correct diagram AND an accurate written response.</i> For a correctly labelled kinked-demand-curve diagram showing the kink at price P^*, the more elastic upper portion (price rises), the less elastic lower portion (price cuts), and the discontinuity in MR AND a clear explanation that rivals match price decreases but ignore price increases, so raising price loses sales sharply and cutting price gains few sales, producing price rigidity within the MR discontinuity [1] — changes in MC within this range leave profit-max P and Q unchanged [1].	3-4
---	---	-----

(iv) Identify the Nash equilibrium. [4]

Check best responses:

- If Sainsbury's chooses High, Tesco's best response: Low ($45 > 40$).
- If Sainsbury's chooses Low, Tesco's best response: Low ($20 > 10$).
- If Tesco chooses High, Sainsbury's best response: Low ($35 > 30$).
- If Tesco chooses Low, Sainsbury's best response: Low ($15 > 8$).

Low is the dominant strategy for both firms. The unique **Nash equilibrium is (Low, Low) with payoffs (£20 m, £15 m)**. This is a prisoner's-dilemma outcome — both firms would be better off choosing (High, High) giving (£40 m, £30 m) — but each has an incentive to deviate.

Identifying (Low, Low) as equilibrium [1]. Recognising dominant strategy reasoning [1]. Correct best-response analysis [1]. Identifying the prisoner's-dilemma property [1].

(v) Distinguish between collusive and non-collusive oligopoly. [2]

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1	Distinction expressed vaguely or only one side identified.
2	In collusive oligopoly firms cooperate formally (cartels) or tacitly (price-leadership) to set prices/output jointly, acting like a monopoly. In non-collusive oligopoly firms act strategically but independently — each firm's decisions depend on anticipated reactions of rivals (e.g. kinked demand, price wars, non-price competition).

(vi) Define the term nudge. [2]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>Vague definition.</i> The idea that a nudge is a way of encouraging people to make certain choices.	1
2	<i>Accurate definition.</i> An understanding that a nudge is a behavioural-economics intervention that alters the choice architecture to influence decisions in a predictable way without restricting options, imposing financial incentives or using coercion.	2

(vii) Explain two reasons why consumer decisions may not be rational. [4]

Marks	Level descriptor
-------	------------------

0	<i>Does not meet standard.</i>
1–2	Two reasons stated vaguely, OR one reason explained clearly. A candidate with one vague reason may be awarded [1].
3–4	Two reasons explained clearly. Possible reasons: bounded rationality (limited time/information/cognitive capacity); anchoring bias (reliance on reference prices e.g. "was £5, now £3"); framing effects (loss aversion framing); availability heuristic; bounded self-control (impulse at checkout); herd behaviour / social proof; default options and status-quo bias.

(viii) State one firm objective alternative to profit maximisation. **[1]**

Any **one** of: revenue maximisation, market-share / sales maximisation, growth maximisation, satisficing, corporate social responsibility (CSR).

(b) Recommend a policy to protect consumers in the UK grocery market. **[10]**

Possible policies:

- Strengthening competition law / merger control by CMA
- Price controls on essential staples (food inflation cap)
- Behavioural regulation — restrictions on checkout placement, BOGOF offers on HFSS foods
- Mandatory unit pricing / clearer labelling
- Supporting the growth of discounters (Aldi/Lidl) to reduce CR4
- Investigating collusive behaviour and imposing fines on cartels
- Nudges — default healthier options, warning labels

Assessment Criteria

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1–2	Little understanding; irrelevant theory; no text/data.
3–5	Some understanding; theory described; superficial synthesis; limited text/data.
6–8	Appropriate theory explained; diagram(s) used; balanced evaluation; text/data applied.
9–10	Fully-developed, balanced, reasoned recommendation using theory, diagram and text/data.

Sample response (abridged): *Given the CR4 of 65%, evidence of collusive potential from game-theoretic analysis (prisoner's dilemma: firms gravitate toward (Low, Low) only under price-war conditions; tacit coordination on high prices is plausible), and consumer behavioural vulnerabilities, I recommend a combined policy: (1) mandatory unit-pricing regulations to counter anchoring; (2) CMA investigations of tacit collusion; and (3) targeted nudges removing HFSS displays at checkouts. Nudges respect consumer autonomy (maintaining choice) while protecting vulnerable consumers. However, price caps risk unintended shortages and reduced supplier margins. A proportionate package of behavioural regulation and antitrust enforcement is likely most effective.*

Question 2 — Bangladesh development & inequality

2. (a) (i) Define the term Gini coefficient. [2]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>Vague definition.</i> The idea that the Gini coefficient measures inequality in a country.	1
2	<i>Accurate definition.</i> An understanding that the Gini coefficient is a numerical measure of income (or wealth) inequality derived from the Lorenz curve, ranging from 0 (perfect equality) to 1 (perfect inequality); it is calculated as the area between the line of perfect equality and the Lorenz curve divided by the total area under the line of perfect equality.	2

(ii) Calculate cumulative share of income received by the poorest 60%. [2]

Cumulative share = 4.0 + 8.0 + 13.0 = **25.0%**.

Valid working [1]. Correct answer with % sign [1].

(iii) Draw and label a Lorenz curve. [3]

The diagram must contain:

- Horizontal axis: cumulative % of households (0→100); vertical axis: cumulative % of income (0→100). [1]
- The line of perfect equality (45° diagonal from (0,0) to (100,100)). [1]
- A correctly plotted Lorenz curve passing through the five cumulative points: (20%, 4%), (40%, 12%), (60%, 25%), (80%, 45%), (100%, 100%), lying below the diagonal. [1]

Candidates who mislabel axes may receive a maximum of [2].

(iv) Outline one cause of income inequality. [2]

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1	<i>Vague outline.</i> States a cause without explanation, e.g. "some people earn more".
2	<i>Accurate outline.</i> Any one of: unequal ownership of factors of production (especially capital/land); unequal access to education; discrimination by gender, caste or ethnicity; regressive tax systems; differences in human capital; rural-urban disparities; monopoly power; globalisation creating skilled-biased returns.

(v) State two HDI indicators. [2]

Any **two** (1 mark each):

- Life expectancy at birth (health).
- Expected years of schooling and mean years of schooling (education).
- GNI per capita (PPP\$) (income / standard of living).

(vi) Calculate % change in HDI 2010–2022. [2]

% change = $((0.670 - 0.500) / 0.500) \times 100 = \mathbf{+34.00\%}$.

Valid working [1]. Correct answer to 2 d.p. with % sign [1].

(vii) Explain, using a PPC or LRAS diagram, how investment in education generates long-run growth. [4]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>There is a correct diagram OR an accurate written response.</i> For a PPC shifting outward OR an LRAS curve shifting right OR an explanation that investment in education raises human capital and labour productivity [1], increasing the productive capacity of the economy [1].	1-2
2	<i>There is a correct diagram AND an accurate written response.</i> For a correctly labelled diagram (PPC shifting out OR LRAS shifting right) AND an explanation that investing in education improves human capital — the skills, knowledge and productivity of the workforce [1]; higher labour productivity and innovation raise the economy's potential output, shifting the PPC outward / LRAS to the right, enabling long-run growth [1].	3-4

(viii) Outline the meaning of microfinance. [2]

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1	<i>Vague outline.</i> The idea that microfinance involves small loans to poor people.
2	<i>Accurate outline.</i> Microfinance is the provision of small-scale financial services — loans, savings, insurance, money transfers — to low-income individuals or groups who are typically excluded from the formal banking sector, intended to support entrepreneurship, smooth consumption and help escape poverty.

(ix) Distinguish between absolute and relative poverty. [2]

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1	Distinction expressed vaguely.
2	Absolute poverty refers to people living below the minimum income necessary to satisfy basic physical needs (e.g. the World Bank threshold of \$2.15/day PPP). Relative poverty refers to a state where income is below a given percentage (commonly 60%) of the median income in a society, indicating exclusion from typical living standards even if basic needs are met.

(b) Recommend a policy to reduce income inequality in Bangladesh. [10]

Possible policies:

- Progressive direct taxation (e.g. a more progressive income tax)
- Expansion of transfer payments / cash-transfer programmes
- Provision of merit goods — public education and healthcare
- Minimum-wage legislation
- Land reform

- Expanding microfinance / access to credit
- Investment in rural infrastructure and agriculture

Assessment Criteria

Marks	Level descriptor
0	<i>Does not meet standard.</i>
1–2	Little understanding; irrelevant theory; no text/data.
3–5	Some understanding; theory described; superficial synthesis; limited text/data.
6–8	Appropriate theory explained; diagram(s) used; balanced evaluation; text/data applied.
9–10	Fully-developed, balanced, reasoned recommendation using theory, diagram and text/data.

Sample response (abridged): *With the richest 20% receiving 55% of income and the poorest 60% only 25%, Bangladesh displays substantial inequality. I recommend expanding public provision of primary and secondary education — a policy that addresses both equity and long-run growth (as (a)(vii) showed, it shifts the LRAS outward). Targeted cash transfers conditional on school attendance would further reduce short-term poverty while breaking inter-generational inequality. Complementary microfinance could reach 98%-repayment borrowers underserved by banks. Trade-offs include fiscal cost (requiring progressive taxation) and potential disincentive effects. On balance, an integrated package of education spending, conditional cash transfers and expanded microfinance is likely most effective, combining equity and growth objectives.*